NoSQL Models and JSON

A ticket can have a requester,
a changing status, and years of events.

Day 1: the data models behind different application needs
Day 2: two JSON designs for the same ticket

Relational ideas we will keep

Fact	Relational representation	Why it matters
A person has an identity	users.user_id is a primary key	Names can change without changing identity
A ticket has a requester	tickets.requester_id refers to users	A relationship should have a defined meaning
A ticket has many events	Each event refers to its ticket	Separate history can grow independently

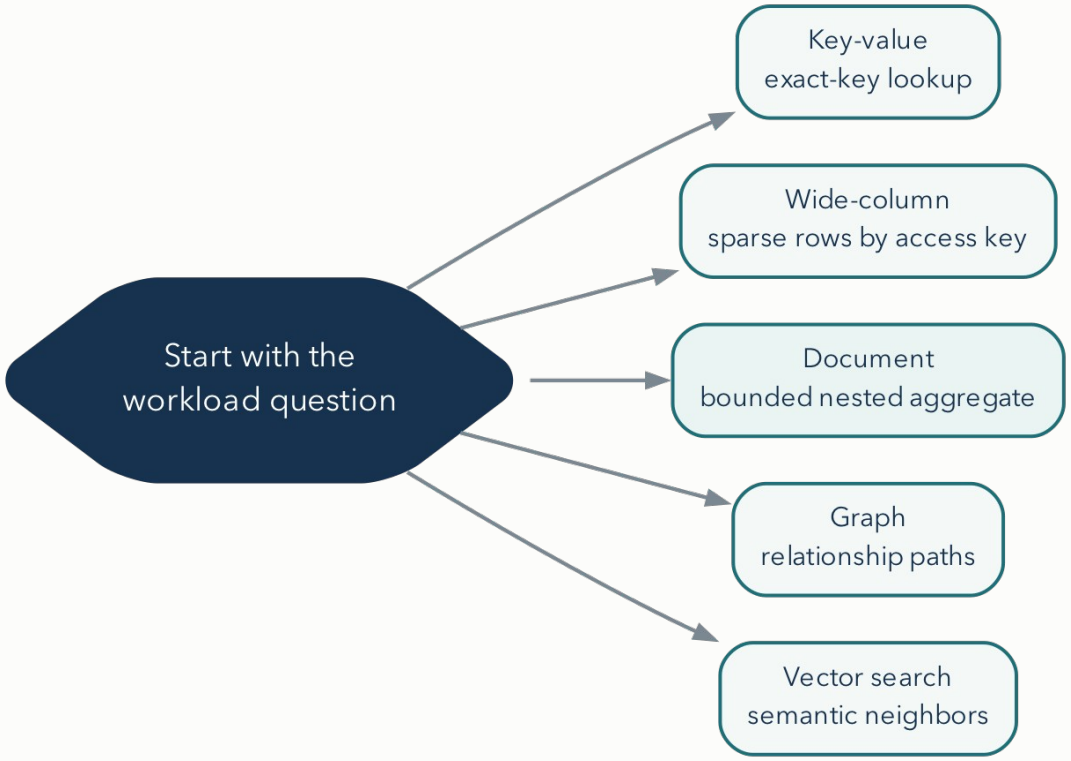
A document design still needs identity, relationships, and rules.
The main change is where related facts live.

Historical roots and distributed workloads

Anchor	Contribution
1970: Codd	Relations and data independence
Earlier NoSQL name	Strozzi used the name for a relational system without SQL
2006: Bigtable	A sparse distributed map for large structured datasets
2007: Dynamo	Key-value storage designed around availability requirements
Later NoSQL movement	A broad label for several nonrelational approaches

These designs address different needs. Dynamo is the research system, not the same product as today's DynamoDB service.

Model families and access patterns



Access pattern: the data an application repeatedly reads or changes.

Opening ticket 1001 and its latest events is one pattern.

Counting every event by type is another.

One application may benefit from more than one representation.

Key-value lookup

session:8f21



Stored session value

```
user_id: 101
expires_at: 2026-10-20T19:30:00Z
```

Known key: fetch this session.

Different request: find every session belonging to user 101.

The second request may need another index or mapping.
The value shown is conceptual notation; it is not a command or JSON.

Wide-column rows and key design

Illustrative row key	Observed fields
device07#2026-10-20T09:00Z	temperature: 21.4
device07#2026-10-20T09:01Z	temperature: 21.6, battery: 78
device12#2026-10-20T09:00Z	humidity: 61

A device-prefix range keeps one device's readings together.
A cross-device battery report has a different access pattern.

Bigtable-style illustration. Wide-column storage differs from analytical columnar storage.

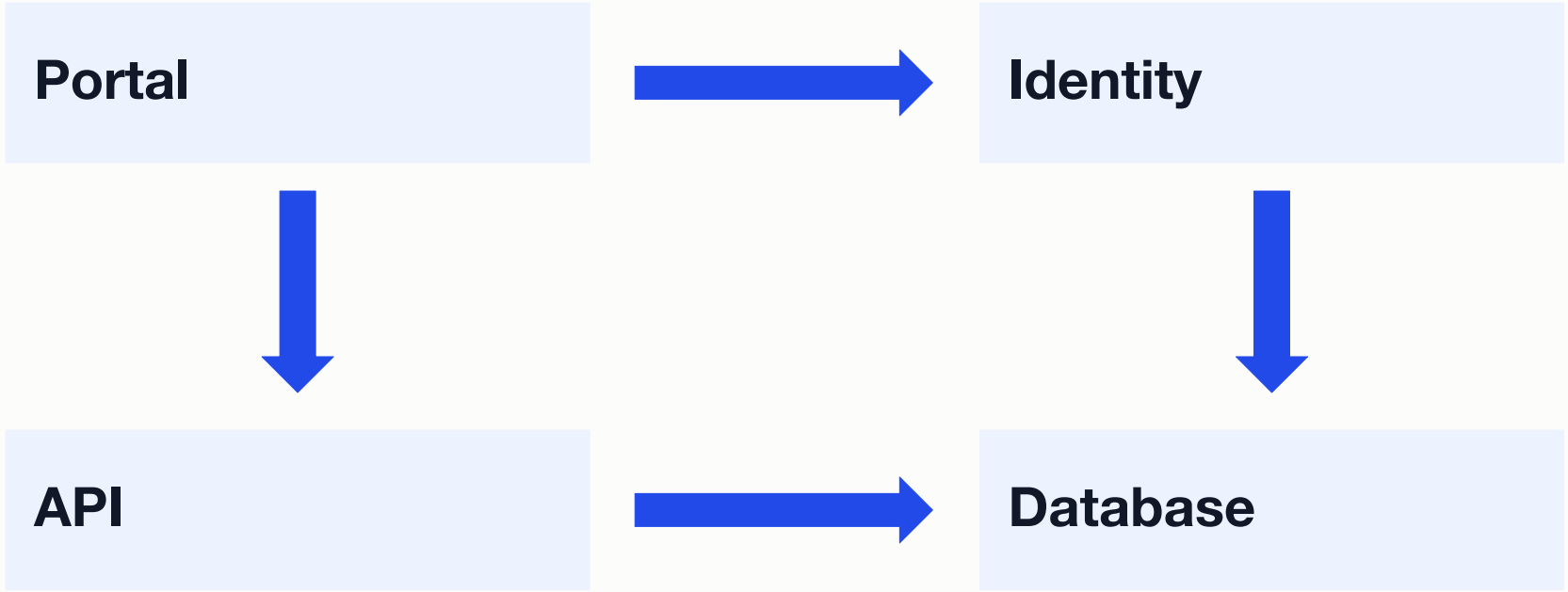
Documents and bounded related data

```
{
  "ticket_id": 1001,
  "status": "open",
  "requester": {"user_id": 101, "display_name": "Requester 101"},
  "tags": ["streetlight", "safety"]
}
```

One document contains the ticket and selected related details.
The nested requester is still a relationship.

Shape preview only. Day 2 teaches the punctuation.
The tags illustrate an added field rather than a column copied from the CSV.

Directed graphs: vertices and edges




```
G = (V, E)
V: four
vertices
E: four edges
```

Each arrow means “depends on.” Portal has two outgoing edges. Portal can reach Database through Identity or API.

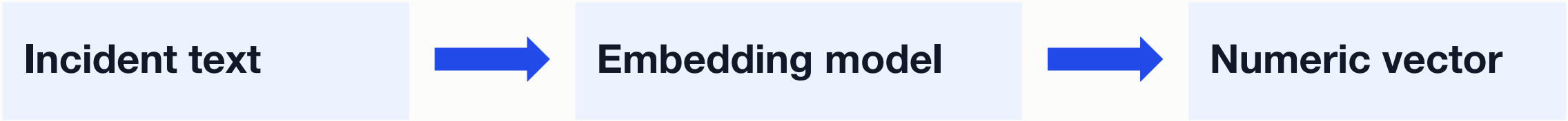
Reachability and breadth-first search

Traversal from Portal	Newly reached vertices
0 edges	Portal
1 edge	Identity, API
2 edges	Database

Breadth-first search visits one edge-distance level at a time.
A visited set prevents repeated visits, including in cycles.

Fewest edges does not mean least latency when edges have different weights.
A dependency path alone does not prove that an outage will propagate.

Vectors and similar incident descriptions

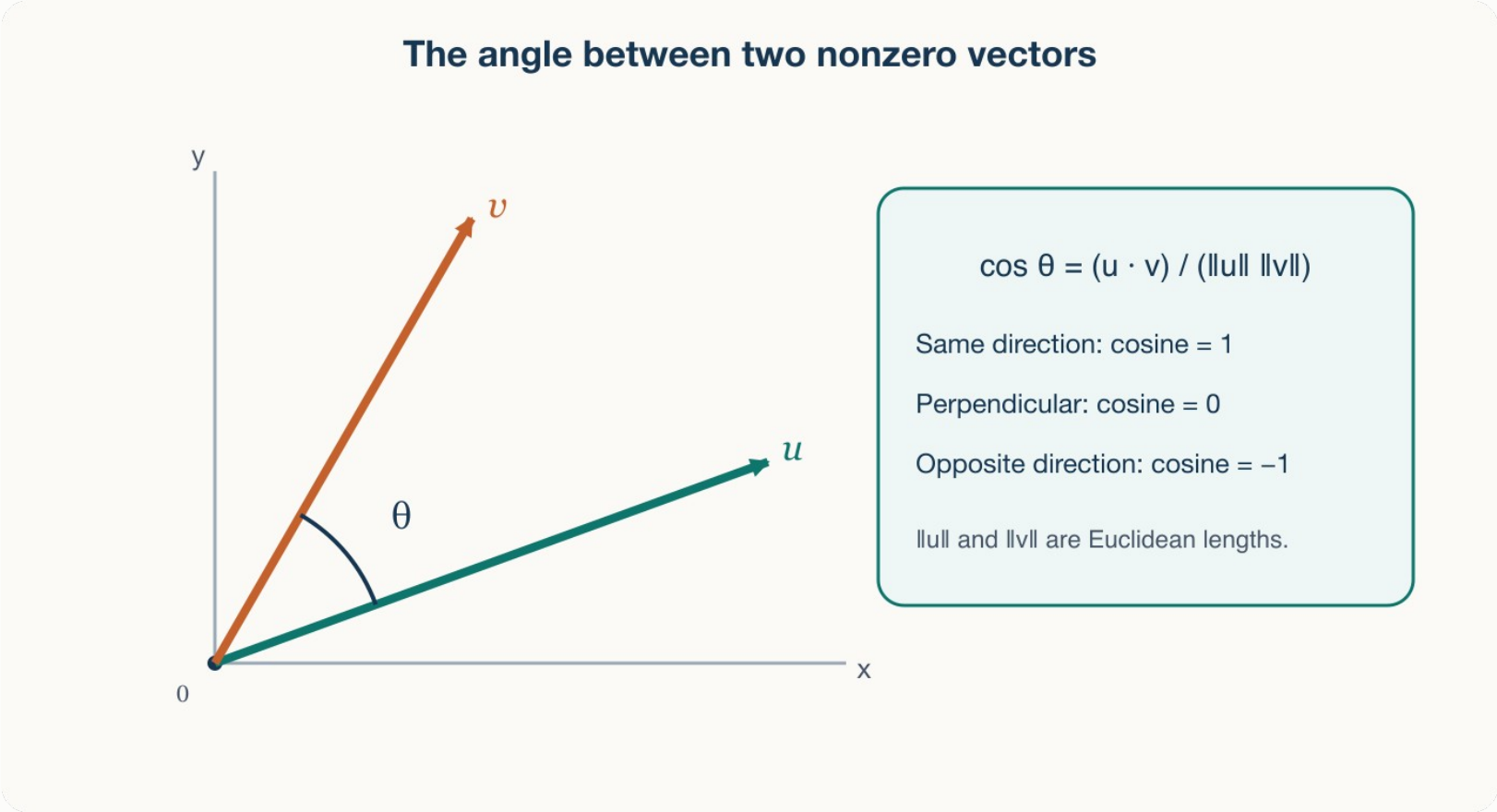


“Lamp outside the library is flickering.”
“Streetlight repeatedly turns off and on.”

A model can place related descriptions near each other.
The database stores the vectors and retrieves nearby candidates.

No embedding API key or vector deployment is required this week.

Vector direction and length



Theta is the angle between u and v. Scaling a vector by a positive factor changes its length but preserves its direction.

Cosine similarity: a complete calculation

```
q = (1, 0)      a = (10, 0)      b = (1, 1)

q · a = 1 × 10 + 0 × 0 = 10
q · b = 1 × 1  + 0 × 1 = 1
```

Lengths: $\|q\| = 1$, $\|a\| = 10$, $\|b\| = \sqrt{2}$

$\cos(q, a) = 10 / (1 \times 10) = 1$
$\cos(q, b) = 1 / (1 \times \sqrt{2}) \approx 0.707$

Cosine ranks a first. A zero-length vector would make the denominator zero.

Euclidean distance reverses this ranking

Distance = √(sum of squared coordinate differences)

```
d(q, a) = √((1 − 10)² + (0 − 0)²) = 9
d(q, b) = √((1 − 1)²  + (0 − 1)²) = 1
```

Euclidean distance ranks b first.
Cosine ignores the positive change in length that makes a farther away.

For unit-length vectors: distance² = 2 − 2 × cosine.
On the same normalized vectors, the exact rankings agree, allowing for ties.

Approximate search and useful results

```
Exact top five IDs:          A, B, C, D, E
Approximate top five IDs:  A, C, E, F, G

Recall at five = 3 / 5 = 0.6
```

Recall measures agreement with the exact retrieval target.
People still have to judge whether the results help with the incident.

Similarity does not grant permission to read a private ticket.
Embedding choice, index settings, and access controls all affect the system.

Model choice for a stated workload

Workload	Candidate	A cost to investigate
Known session token	Key-value	Finding sessions without their keys
One device's ordered readings	Wide-column	Other query patterns and hot keys
One ticket with bounded details	Document	Shared facts and growing arrays
Multi-hop dependencies	Graph	Traversal size and edge meaning
Similar incident descriptions	Vector search	Retrieval quality and model versions
Shared facts and changing reports	Relational	Joins and schema evolution

These are starting points. Products often support more than one model.

A short design exercise

The support portal needs a ticket page and a report of event counts.

Choose a useful representation for the ticket page.
Then explain one consequence for the report or for changing a name.

Work individually in your notes. No separate submission.
Keep the relational option available and justify any additional store.

Day 2: JSON and the ticket page

The same facts can appear
in different document shapes.

Syntax and types, followed by two worked designs
GitHub editor and Atlas orientation
Individual CSV-to-JSON lab

JSON as a language-independent format

2006: RFC 4627 documents JSON
2014: RFC 7159 revises interoperability rules
2017: RFC 8259 becomes the Internet standard

```
{"ticket_id": 1001, "status": "open"}
```

JavaScript, Python, and other languages can exchange the same text.
Objects and arrays directly express nested API responses.

JSON describes data. It does not execute a query or provide a database.

JSON has six value kinds

Kind	Example	Meaning
Object	{"status": "open"}	Named members
Array	["streetlight", "safety"]	Ordered values
String	"open"	Quoted text
Number	1001	Numeric value
Boolean	true or false	Two literals in one value kind
Null	null	An explicit null value

Any JSON value can be the top level. Our lab deliberately uses an object.

Objects, arrays, and punctuation

```
{
  "ticket_id": 1004,
  "subject": "Broken bench slat",
  "status": "new",
  "assignee_id": null,
  "tags": ["parks", "bench"]
}
```

{ } enclose an object

[ ] enclose an array

: connects name and value

, separates neighbors

Names and strings use double quotes. Indentation shows containment.
The tags are an illustrative addition to ticket 1004.

A syntax repair

```
{'status': 'open',}
```

Two errors: single quotes and a comma after the final member.

```
{"status": "open"}
```

Comments, NaN, Infinity, and the number 01 are also outside standard JSON.
A permissive parser may accept extensions. Acceptance alone is insufficient.

Missing, null, empty, and zero

JSON fragment	What the text supplies
{}	No assignee property
{"assignee_id": null}	An explicit null assignee
{"events": []}	An empty array
{"estimated_cost": 0}	A numeric zero

The application decides whether an omitted field and null mean the same thing.
An empty event array does not explain why history is absent.

JSON text and MongoDB BSON

```
{"opened_at": "2026-02-01T23:10:00Z"}
```

This value is a JSON string with an agreed timestamp format.

BSON supports additional storage types, including Date and ObjectId.
A driver or an import step decides how to create those stored types.

Use unique member names. Keep IDs as strings when their exact spelling matters.
A valid document can still violate the application's data contract.

The worked example uses ticket 1001

CSV	Relevant records
users.csv	101: Requester 101. 201: Agent 201.
tickets.csv	1001: Streetlight dark near bus stop. Status: open. Requester: 101. Assignee: 201.
ticket_events.csv	5001: created, 2026-02-01T23:10:00Z 5002: assigned, 2026-02-02T14:05:00Z

Both upcoming examples preserve these facts. Other CSV columns are omitted. The student lab uses a different ticket, 1003, with three events.

Referenced design: IDs connect separate records

```
{
  "users": [
    {"user_id": 101, "display_name": "Requester 101"},
    {"user_id": 201, "display_name": "Agent 201"}
  ],
  "tickets": [{
    "ticket_id": 1001, "subject": "Streetlight dark near bus stop",
    "status": "open", "requester_id": 101, "assignee_id": 201
  }],
  "events": [
    {"event_id": 5001, "ticket_id": 1001, "event_type": "created",
     "event_at": "2026-02-01T23:10:00Z"},
    {"event_id": 5002, "ticket_id": 1001, "event_type": "assigned",
     "event_at": "2026-02-02T14:05:00Z"}
  ]
}
```

One JSON object packages three arrays. This file is not yet a MongoDB database.

Embedded design: related facts inside the ticket

```
{
  "ticket_id": 1001,
  "subject": "Streetlight dark near bus stop",
  "status": "open",
  "requester": {"user_id": 101, "display_name": "Requester 101"},
  "assignee": {"user_id": 201, "display_name": "Agent 201"},
  "events": [
    {"event_id": 5001, "event_type": "created",
     "event_at": "2026-02-01T23:10:00Z"},
    {"event_id": 5002, "event_type": "assigned",
     "event_at": "2026-02-02T14:05:00Z"}
  ]
}
```

The containing ticket identifies the events' owner.
The document carries copies of the two names.

A current name and a historical name

Meaning of the copied name	Consequence when Requester 101 changes it
Current display name	Every current-name copy must eventually match the authoritative user record
Name when the ticket opened	Keep the old name deliberately and label it as a snapshot
User ID without a copied name	Look up the authoritative user record for the current name

A stale current name is an inconsistency.
A documented historical snapshot preserves a different fact.

Bounded recent events and complete history

Ticket document	Separate event records
Ticket ID and status	ticket_id links each event
At most five recent events	All retained events
Fast ticket-page preview	Independent history and reports

A copied recent-event list needs a refresh rule.
This design does not make changes across separate documents automatically atomic.

GitHub's browser text editor

Your repository: Add file > Create new file
Filename: week_09_json_models.md

````
## Referenced design
```json
{"ticket_id": 1004, "assignee_id": null}
```

My explanation goes below the JSON block.
````

Preview checks the Markdown display. Commit changes saves the text.
Download the file for Brightspace. Neither step validates the JSON syntax.

Atlas orientation and connection boundaries

Component	Purpose
Atlas account and project	Manage the cloud deployment
Free cluster	Run the database service with free-tier limits
Database user	Authenticate a database connection
IP access list	Permit a source network to attempt that connection

Choose Free / M0 and only the access needed for your practice.
Do not enter a payment card or publish credentials.

Today's JSON lab needs no cloud connection. Week 10 introduces MQL.

Lab: one ticket, two JSON designs

Ticket 1003. Requester 103. Assignee 201.
Events 5005, 5006, and 5007 from the Metro Support CSVs.

Write one referenced design and one embedded or hybrid design.

Explain how the ticket page reads the facts and how name changes and years of history affect your choice.

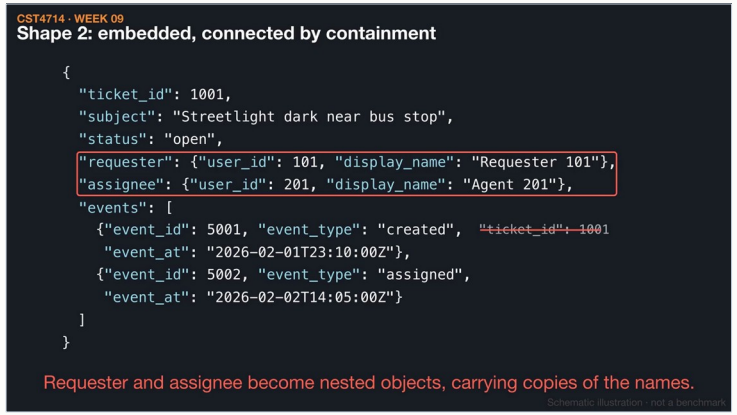
Individual work in a text editor. No SQL, MQL, or Python required.
Brightspace: one Markdown file containing both examples and the paragraph.

Explaining a document-design decision

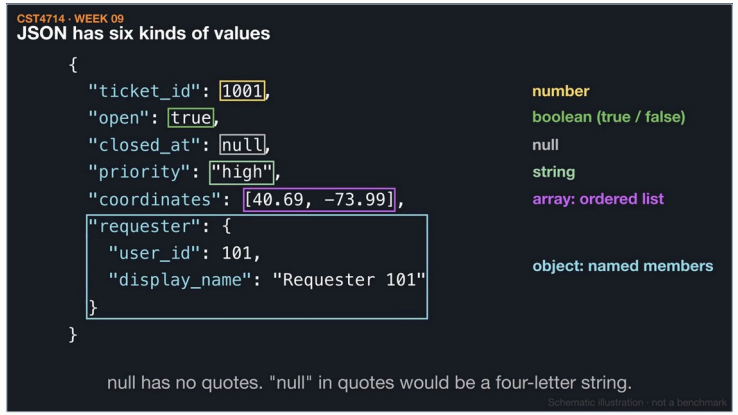
I kept user IDs so a name change would not change identity. For the ticket page, I considered embedding a small set of recent events. I would keep the complete history separately because it continues to grow. A recent-event copy would need a refresh rule so readers could distinguish an up-to-date preview from an older one.

The lab practices this reasoning with concrete JSON.
Next week: retrieve and change documents with MQL.

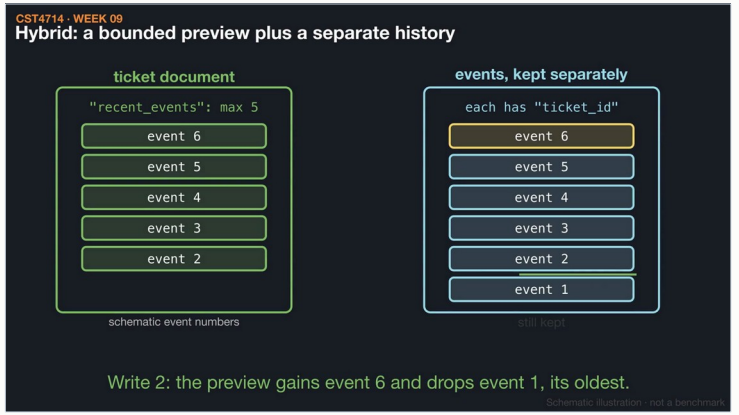
This week's videos



▶ Same facts, two shapes (1:16)



▶ Strict JSON, from the punctuation up (1:20)



▶ What embedding costs over time (1:21)

In PowerPoint Slide Show, click a thumbnail to play full screen. Embedded videos work offline.